

ENGINEERING DECISION-SUPPORT ASSESSMENT - This is not an insurance instrument, credit/lending document, or guarantee of future performance. Requires review and sign-off by a certified reliability engineer before any action.

Overall Assessment

51

Overall health score (0-100)
Classification: REQUIRES ATTENTION | Risk class: HIGH

1. Asset Details

Turbine	WT-010 (asset ref 10)
OEM / Model	GE 2.5XL
Rated power	2500 kW
Site / Location	Wind Farm A
Commissioned	Not recorded (recommend adding to asset master)
Assessment date	2026-06-02
Valid until	2026-08-31 (90-day validity)

2. Data Provenance

SCADA coverage	100.0% of assessed window has complete key signals
Rows assessed	1,000 records (10-minute averages)
Sampling note	10-min averaged SCADA hides high-frequency early signatures; high-frequency/vibration data recommended for earlier warning.

3. Model Traceability

Model version	v2026.05.31.1247
Training window	xgboost v2026.05.31.1247, balanced sample 2450 rows, trained 2026-05-31
Calibration	Calibrated (isotonic) - ECE 0.3881->0.114, Brier 0.4308->0.2677
Fault probability	49% (calibrated), component: Gearbox

4. Component Scores & Remaining Useful Life

Component	Health (0-100)	RUL estimate (with range)
Gearbox	20	~8 days (range 5-10 d)
Generator	99	~29 days (range 20-37 d)
Main Bearing	46	~15 days (range 10-19 d)
Hydraulic	100	~21 days (range 14-27 d)
Transformer	100	~26 days (range 18-33 d)
Pitch	46	~15 days (range 10-19 d)

5. Plain-language Assessment

WT-010 presents an overall asset health score of 51/100, classified as REQUIRES ATTENTION. The WindSense ML model (v2026.05.31.1247) estimates a 49% calibrated probability of an anomaly, with the Gearbox subsystem as the leading driver. Estimated Remaining Useful Life for the affected component is 17 days (indicative). Leading SHAP contributors: gen_bearing_DE_margin (24.6%), hydraulic_temp_margin (16.4%), Temp_hydraulic_oil_group_C (12.9%). This is an engineering decision-support assessment generated from normalised SCADA data; it is not an insurance instrument or a guarantee of future performance.

6. Limitations & Known Miss-Rate

Developed on the CARE open benchmark: 5 turbines, 11 labelled failure events, 10-minute averaged SCADA (hides early high-frequency signatures). Forward-in-time (purged/embargoed) PR-AUC ~ 0.41 - close to base rate, so prediction of UNSEEN FUTURE failures is NOT yet validated. KNOWN MISS RATE: high-severity failure-mode downgrade rate $\sim 100\%$ (gearbox/generator failures may be classified as a lower-severity fault); component attribution is INDICATIVE ONLY. This document is engineering decision-support, not a validated insurance or lending instrument.

7. Certifying Engineer

Engineer

A.Khan RE-2291

Signature

Date / License no.